

CHRISTOPHER J. EGAN

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Robotics Engineering and Computer Science junior at Worcester Polytechnic Institute specializing in computer vision, state estimation and machine learning. Experience in developing perception and control systems across research, embedded systems and over three years of international rocketry and robotics competitions

SKILLS & TECHNOLOGIES

Languages: C, C++, Python, Java, MATLAB, TypeScript, HTML, CSS, Racket

Software/Frameworks: Linux, Ubuntu, ROS, ROS2, Autodesk Inventor, OpenCV, CUDA, PyTorch, gcc/gcc++, Simulink, PlatformIO, git, Nerfstudio, Claude Code

Hardware: STM32H7, ESP32, Raspberry Pi, romi32U4, Realsense, Vicon, NVIDIA Jetson, LiDAR

EDUCATION

Worcester Polytechnic Institute (WPI)

2024 - 2028

B.S. Robotics Engineering & B.S. Computer Science (Dual Degree)

- Notable Courses: Object Oriented Design, Differential Equations, Algorithms, Operating Systems
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EXPERIENCE

Perception & Machine Learning Engineer

Feb. 2026 - Present

PeAR (Perception and Autonomous Robotics Lab)

Worcester, MA

- Researched and implemented optimal ways to get high quality, metric Gaussian splats fusing different combinations of tools such as Vicon, LiDAR, RGB images, depth images and IMUs
- Designed a pipeline to go from a rendered metric Gaussian splat to a Euclidean Signed Distance Field
- Contributed to an award finalist MQP by integrating the ESDF pipeline with a dual-quaternion NMPC to guide a quadcopter.
- Integrated Gaussian splat and ESDF into reinforcement learning platform, generating photo-realistic hallucinated depth images & event camera images as observations to train drones to avoid obstacles using Gaussian splat generated ESDF as collision detection.

Research Engineer

Aug. 2026 - Present

FuTURE (The Future Therapeutic Unlimited Robotics Engineering Lab)

Worcester, MA

- Selected to contribute to development of AI assistant for endoscopy and stomach image reconstruction in collaboration with Boston Scientific.
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Leadership & Projects

Guidance, Navigation & Control Lead

Aug. 2025 - Present

WPI High Powered Rocketry Club

Worcester, MA

- Researched and Developed a quaternion-based multiplicative extended Kalman filter (QM-EKF) that fused dual IMUs, GPS, barometer, and magnetometer measurements to estimate full 6-DoF vehicle state for autonomous rocket guidance in a MATLAB/Simulink flight simulation
- Ported QM-EKF to C++ on an STM32H7 microcontroller for a planned supersonic flight with an anticipated apogee of 10,000 feet, using canards for stabilization with poses from the QM-EKF
- Developed software to interface with two IMUs on an STM32H7 and collect 7 hours of synchronized data and run an Allan variance test for the QM-EKF's noise matrix